Linear Algebra and Its Applications

Section 6.8: Applications of Inner Product Spaces

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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6.8 Applications of inner product spaces

Two conventions matter. The book uses W to multiply *residuals*, so weighted least squares minimizes $\|W(y - X\beta)\|^2$. Therefore its normal matrix contains $W^TW = W^2$ when W is diagonal. Do not replace this by W unless the supplied weights have first been square-rooted. For Fourier approximations, the inner product is the ordinary integral $\langle f, g\rangle = \int_0^{2\pi} fg$; the constant coefficient is $a_0/2$, with $a_0 = (1/\pi)\int f$, while $a_k = (1/\pi)\int f\cos kt$ and $b_k = (1/\pi)\int f\sin kt$. These differing denominators arise because $\|1\|^2 = 2\pi$ and $\|\sin kt\|^2 = \|\cos kt\|^2 = \pi$ for positive integer k.

Practice problem 6.8.P1: Check the three integral products

For $q_1 = 1$, $q_2 = t$, $q_3 = 3t^2 - 4$ on $[-2, 2]$,

$$\langle q_1, q_2\rangle = \int_{-2}^{2} t\,dt = 0,$$

$$\langle q_1, q_3\rangle = [t^3 - 4t]_{-2}^{2} = 0,$$

$$\langle q_2, q_3\rangle = \int_{-2}^{2} (3t^3 - 4t)\,dt = 0.$$

The first and third vanish also by oddness. All three polynomials are nonzero, so the pairwise zero products prove an orthogonal set.

Practice problem 6.8.P2: Read approximations from a finite trigonometric expression

The function is already $f(t) = 3 - 2\sin t + 5\sin 2t - 6\cos 2t$. The first-order space retains only the constant and frequency-one terms, so its approximation is $3 - 2\sin t$. The third-order space contains all the displayed terms (there are no frequency-three terms to add), so its approximation is f itself. Orthogonality makes the discarded higher-frequency terms perpendicular to the lower-order space; this is why simply retaining terms gives the least-squares approximation here.

Exercise 6.8.1: Half-size residual multipliers at the two endpoints

Use design rows $(1, x_i)$ for $x_i = -2, -1, 0, 1, 2$ and $y = (0, 0, 2, 4, 4)^T$. Following the book's convention, take $W = \mathrm{diag}(1, 2, 2, 2, 1)$: the endpoint residuals receive half the multiplier of the interior ones. Scaling all multipliers by a common positive factor does not change the fit. The weighted normal equations are

$$X^TW^2X\widehat{\beta} = X^TW^2y, \qquad \begin{bmatrix} 14 & 0 \\ 0 & 16 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 28 \\ 24 \end{bmatrix}.$$

Thus $\widehat{\beta}_0 = 2$ and $\widehat{\beta}_1 = 3/2$.

Conclusion. The fitted line is $y = 2 + (3/2)x$. Here "half" refers to the residual multiplier used in the text, whose square enters the objective.

Exercise 6.8.2: A common rescaling leaves the minimizers unchanged

The second weighting matrix is $W_2 = 2W_1$, because every point's multiplier is doubled. For any parameter vector,

$$\|W_2(y - X\beta)\|^2 = 4\|W_1(y - X\beta)\|^2.$$

Multiplication by the positive constant 4 preserves every ordering of objective values, so the minimizing parameter vectors are exactly the same. The numerical minimum changes by that factor, but the fitted curve does not. This also holds when the minimizer is not unique.

Exercise 6.8.3: Add the cubic trend to the quadratic trend

Use nodes $-2, -1, 0, 1, 2$ and observations $g = (3, 5, 5, 4, 3)$. The orthogonal polynomials from Example 2 and the question are $p_0 = 1, p_1 = t, p_2 = t^2 - 2, p_3 = (5t^3 - 17t)/6$. Their inner products with the data and squared norms are

j	0	1	2	3
$\langle g, p_j \rangle$	20	-1	-7	2
$\langle p_j, p_j \rangle$	5	10	14	10

The cubic values are $(-1, 2, 0, -2, 1)$, so its coefficient is $2/10 = 1/5$. The fitted cubic trend is

$$\widehat{g} = 4p_0 - \tfrac{1}{10}p_1 - \tfrac{1}{2}p_2 + \tfrac{1}{5}p_3 = \boxed{5 - \tfrac{2}{3}t - \tfrac{1}{2}t^2 + \tfrac{1}{6}t^3}.$$

Because the basis is orthogonal, adding the cubic term does not require recomputing the earlier three coefficients.

Exercise 6.8.4: Orthogonal trends at six evenly spaced nodes

(a) Check the evaluation vectors.

At $-5, -3, -1, 1, 3, 5$, the proposed polynomials have lists

$$p_0 : (1, 1, 1, 1, 1, 1), \quad p_1 : (-5, -3, -1, 1, 3, 5),$$

$$p_2 : (5, -1, -4, -4, -1, 5).$$

The lists for p_1 and p_2 each sum to zero, so both are orthogonal to p_0. Their mutual product sums to zero by odd-even cancellation. Their squared norms are $6, 70, 84$, all positive. Thus they form the required orthogonal basis of the quadratic space.

(b) Project the actual data onto that basis.

For $g = (1, 1, 4, 4, 6, 8)$, the three inner products are $24, 50, 6$. Hence

$$\widehat{g} = \frac{24}{6}p_0 + \frac{50}{70}p_1 + \frac{6}{84}p_2 = 4 + \frac{5}{7}t + \frac{1}{14}\left(\frac{3}{8}t^2 - \frac{35}{8}\right).$$

Equivalently the quadratic is $\boxed{\widehat{g}(t) = 59/16 + (5/7)t + (3/112)t^2}$. The projection residual is orthogonal to all three basis lists, verifying the fit.

Exercise 6.8.5: Sines of distinct integer frequencies are orthogonal

For unequal positive integers m, n, use $2\sin mt \sin nt = \cos((m-n)t) - \cos((m+n)t)$. Both $m-n$ and $m+n$ are nonzero integers. Their cosine integrals on $[0, 2\pi]$ are zero, since $\int_0^{2\pi} \cos(kt)\,dt = [\sin(kt)/k]_0^{2\pi} = 0$. Thus the integral of the original product is zero, proving orthogonality.

Exercise 6.8.6: Sine and cosine terms are orthogonal even at equal frequency

Use $2\sin mt \cos nt = \sin((m+n)t) + \sin((m-n)t)$. Every nonzero integer-frequency sine integrates to zero over $[0, 2\pi]$. If $m = n$, the second term is identically $\sin 0 = 0$, so it also has zero integral without any division by zero. Thus $\langle \sin mt, \cos nt \rangle = 0$ for all the stated positive integers.

Exercise 6.8.7: The Fourier basis elements have squared norm pi

For positive integer k, the identities $\cos^2 kt = (1 + \cos 2kt)/2$ and $\sin^2 kt = (1 - \cos 2kt)/2$ give

$$\|\cos kt\|^2 = \tfrac{1}{2}(2\pi) + 0 = \pi, \qquad \|\sin kt\|^2 = \tfrac{1}{2}(2\pi) - 0 = \pi.$$

The Fourier setting uses integer frequencies. For an arbitrary noninteger real k, the boundary term $\sin(4\pi k)/(4k)$ need not vanish, so the integer-frequency condition is what justifies these equalities.

Exercise 6.8.8: A constant shift changes only the constant coefficient

For $f(t) = t - 1$, the mean coefficient is $(1/(2\pi)) \int_0^{2\pi} (t-1)\,dt = \pi - 1$. For $k \geq 1$, integration by parts gives $a_k = 0$ and $b_k = -2/k$; the constant -1 has zero product with those sines and cosines.

Conclusion. The third-order approximation is $\pi - 1 - 2\sin t - \sin 2t - (2/3)\sin 3t$.

Exercise 6.8.9: Reverse the slope of the linear function

For $f(t) = 2\pi - t$, the constant coefficient is π. The nonconstant Fourier coefficients of t have their signs reversed: $a_k = 0$, $b_k = 2/k$. These also follow directly by integrating $(2\pi - t)\sin kt$ by parts.

Conclusion. The third-order approximation is $\pi + 2\sin t + \sin 2t + (2/3)\sin 3t$.

Exercise 6.8.10: Project an integrable square wave onto continuous trigonometric functions

The square wave in the exercise has a jump at π, so it is not literally a member of $C[0, 2\pi]$. Nevertheless it is bounded and Riemann integrable, as are all its products with the finite continuous trigonometric basis. The finite-dimensional least-squares problem $\min_g \int_0^{2\pi} (f-g)^2$ is therefore well defined on that continuous trigonometric subspace. Its normal equations use the same integrals; no assertion about an inner product on all pointwise piecewise-continuous functions is needed. Values assigned at isolated jump endpoints do not change any of these integrals. The mean is zero. For $k \geq 1$,

$$a_k = \frac{1}{\pi}\left(\int_0^{\pi} \cos kt\,dt - \int_{\pi}^{2\pi} \cos kt\,dt\right) = 0,$$

$$b_k = \frac{1}{\pi}\left(\int_0^{\pi} \sin kt\,dt - \int_{\pi}^{2\pi} \sin kt\,dt\right) = \frac{2(1-(-1)^k)}{\pi k}.$$

Thus even sine coefficients vanish and odd ones equal $4/(\pi k)$.

Conclusion. The third-order approximation is $(4/\pi)(\sin t + (1/3)\sin 3t)$.

Exercise 6.8.11: An exact finite trigonometric identity

The identity $\sin^2 t = (1-\cos 2t)/2$ already expresses the function in the third-order target subspace. It is therefore its own best approximation, with zero error. Its only nonzero coefficients are the constant $1/2$ and the cosine-two coefficient $-1/2$.

Conclusion. $1/2 - (1/2)\cos 2t$. No integration is necessary.

Exercise 6.8.12: Use the cosine triple-angle identity

From $\cos 3t = 4\cos^3 t - 3\cos t$, solve for the cube: $\cos^3 t = (3/4)\cos t + (1/4)\cos 3t$. This lies entirely in the third-order target space, so the best approximation is exact, again with zero error.

Conclusion. $(3/4)\cos t + (1/4)\cos 3t$.

Exercise 6.8.13: Each coefficient depends linearly on the function

For example,

$$a_k(f+g) = \frac{1}{\pi}\int_0^{2\pi} (f+g)\cos kt = a_k(f) + a_k(g)$$

by linearity of the integral. Replacing cosine by sine proves the same identity for b_k, and taking the constant term proves it for a_0 too. The fixed denominators are the same for both functions, so no extra normalization appears when adding coefficients.

Exercise 6.8.14: One additional correct Fourier coefficient cannot worsen the fit

The difference is $h - g = b_2 \sin 2t$. Since b_2 is the Fourier coefficient of f, the residual $f - h$ is orthogonal to $\sin 2t$. Therefore the Pythagorean theorem gives

$$\|f-g\|^2 = \|(f-h) + b_2 \sin 2t\|^2 = \|f-h\|^2 + \pi b_2^2.$$

Conclusion. h is at least as close as g. It is strictly closer if $b_2 \neq 0$; if $b_2 = 0$, the two functions and their errors are identical.

Exercise 6.8.15: A weighted cubic takeoff fit

Use the thirteen times and position observations from Exercise 6.6.13, with design row $(1, t, t^2, t^3)$ and the stated diagonal residual multipliers $1, 1, 1, 0.9, 0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1$. The normal equations must use $X^T W^2 X$, not $X^T W X$. For reproducibility their coefficient matrix and right side

are

$$\begin{bmatrix} 6.66 & 22.23 & 120.77 & 797.19 \\ 22.23 & 120.77 & 797.19 & 5956.13 \\ 120.77 & 797.19 & 5956.13 & 48490.23 \\ 797.19 & 5956.13 & 48490.23 & 420477.17 \end{bmatrix} \widehat{\beta} = \begin{bmatrix} 747.844 \\ 4815.438 \\ 35420.468 \\ 285262.44 \end{bmatrix}.$$

Solving with the supplied decimals treated as exact rational data gives, after rounding for display,

$$y(t) \approx -0.2685118943 + 3.6095481671t + 5.8575839604t^2 - 0.04770303371t^3.$$

The exact fitted residual satisfies $X^T W^2 (y - X\widehat{\beta}) = 0$, which checks that this curve minimizes the weighted objective. Differentiate and evaluate at 4.5 seconds:

$$y'(4.5) = \widehat{\beta}_1 + 9\widehat{\beta}_2 + 60.75\widehat{\beta}_3 \approx \boxed{53.42984451 \text{ feet/second}}.$$

The later observations receive less influence because their residuals are multiplied by smaller numbers before squaring.

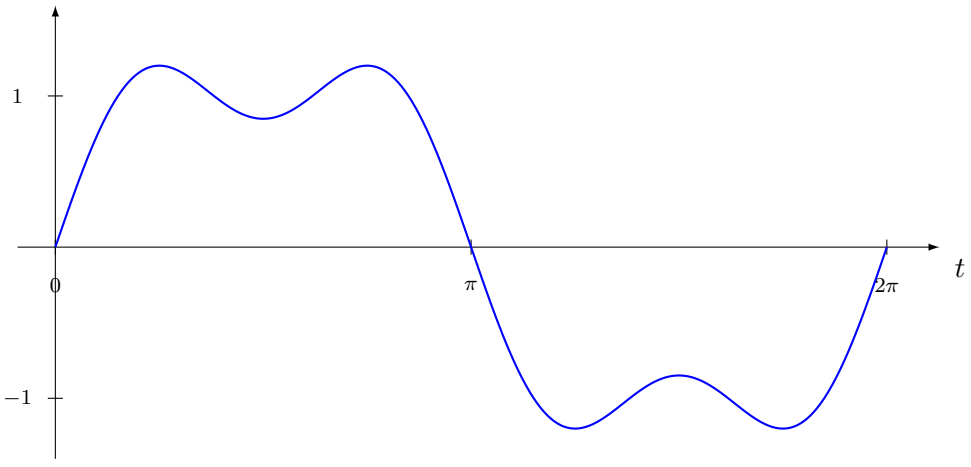
Exercise 6.8.16: Graph two Fourier orders and their periodic continuation

From Exercise 10, even sine coefficients vanish, so

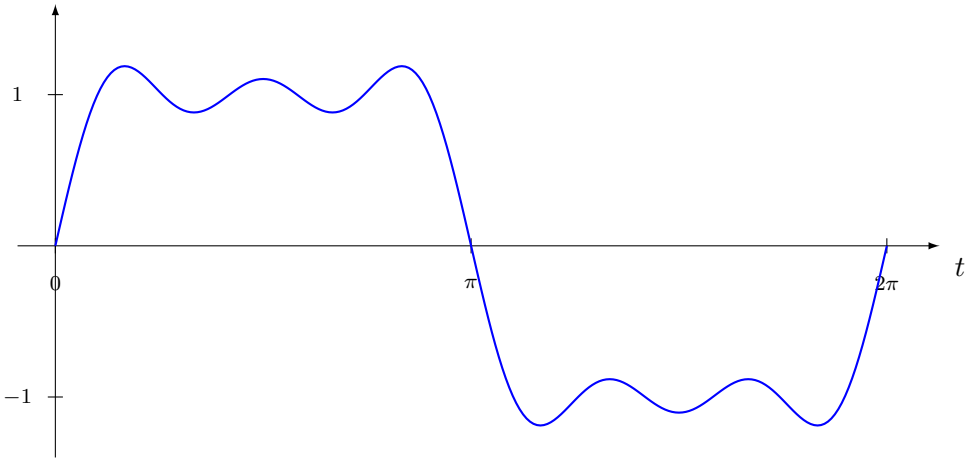
$$f_4(t) = \frac{4}{\pi}\left(\sin t + \frac{\sin 3t}{3}\right), \qquad f_5(t) = f_4(t) + \frac{4}{5\pi}\sin 5t.$$

Thus the fourth-order projection equals the third-order projection: its new frequency-four coefficients are zero. The fifth adds a nonzero frequency-five term. The requested three separate plots follow; the formulas determine the values exactly, while the curves are sampled densely for drawing. Both are 2π-periodic, explaining the continuation on the third plot.

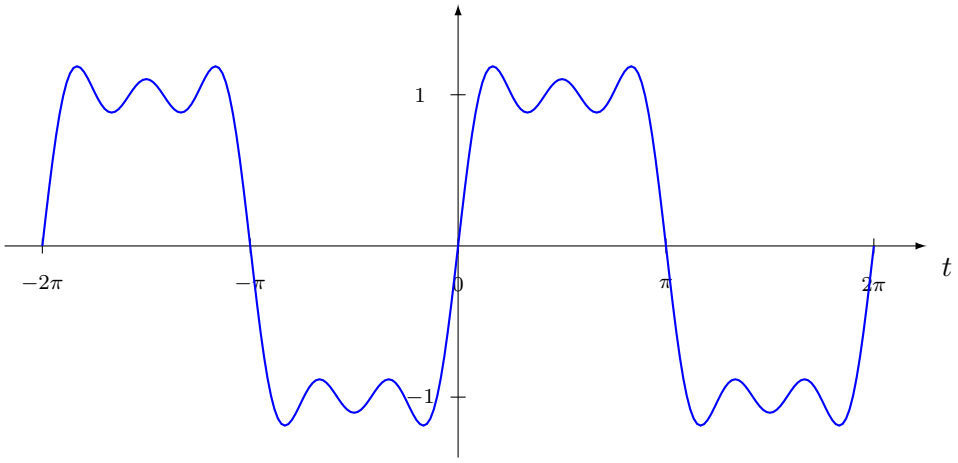
f_4 **on** $[0, 2\pi]$



f_5 **on** $[0, 2\pi]$



f_5 **on** $[-2\pi, 2\pi]$



Near the target's jumps the approximations overshoot; adding terms improves the integral fit without making these low-order continuous curves equal a discontinuous square wave. The finite trigonometric sums have value zero at the jump locations, the midpoint of the adjacent target levels.